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# Hybrid Anomaly Detection System for Stock Market Manipulation Identification

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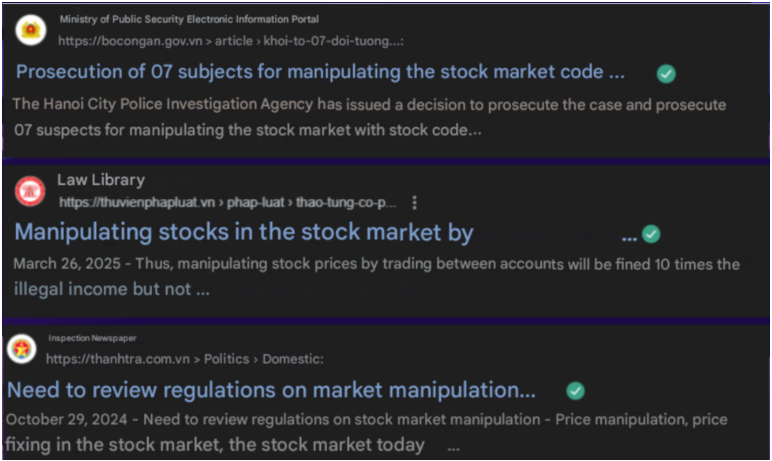
# Outline

1. Methods
2. Performance Evaluation
3. Comparison & Discussion
4. Conclusion

# Practical information related to the project



# Project Goal

**The goal of the project is to find stocks that are at risk of being manipulated.**

**What is it used for?**

- Detect unusual transactions early (e.g., stocks being artificially pushed up or down, manipulated)
- Warm investors of risks before they buy
- Help users know which stocks are volatile
- Support financial research and market behavior

# Research Objectives

- **Our Solution:** A Hybrid Anomaly Detection System
- **Goal:** Create a balanced system between Accuracy, Speed, and Explainability.
- **Core Technology:** Expert Rules (Domain Knowledge) + Isolation Forest (Unsupervised Machine learning)
- **Target:** Real-time alerts for retail investors on standard hardware

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# Hybrid Anomaly Detection System (Rule-based & ML)

## 1. Rule-based Component (Financial Heuristics)

Using quantitative indicators:

- **Daily Return (Tsay, 2010):** $return_t = \frac{Close_t - Close_{t-1}}{Close_{t-1}}$
- **RSI Indicator (Wilder, 1978):** $RSI = 100 - \frac{100}{1+RS}$
- **Bollinger Bands:** $BB_{up/low} = MA_{20} \pm 2\sigma_{20}$

**Activation:** Volume ratio $> 1.8$ and price breakout.

## 2. ML Component (Isolation Forest)

Capturing non-linear patterns:

- **Mechanism:** Anomalies are easier to isolate using tree structures.
- **Input (10D):** Log-return, Volatility, RSI, MACD, ATR, etc.
- **Advantage:** Unsupervised learning.

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# Hybrid Anomaly Detection System (Hybrid Scoring)

## Hybrid Scoring Formula

$$Score_{final} = 0.6 \times Score_{Rule} + 0.4 \times Score_{ML}$$

### 0.6 Rule-based Component

- **Priority:** Interpretability and financial logic.
- **Logic:** Leverages established domain heuristics.

### 0.4 ML-based Component

- **Priority:** Capturing complex non-linear patterns.
- **Logic:** Enhances detection with unsupervised learning.

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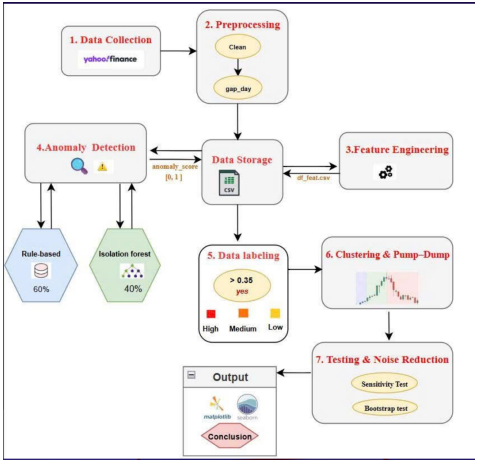
# System Architecture



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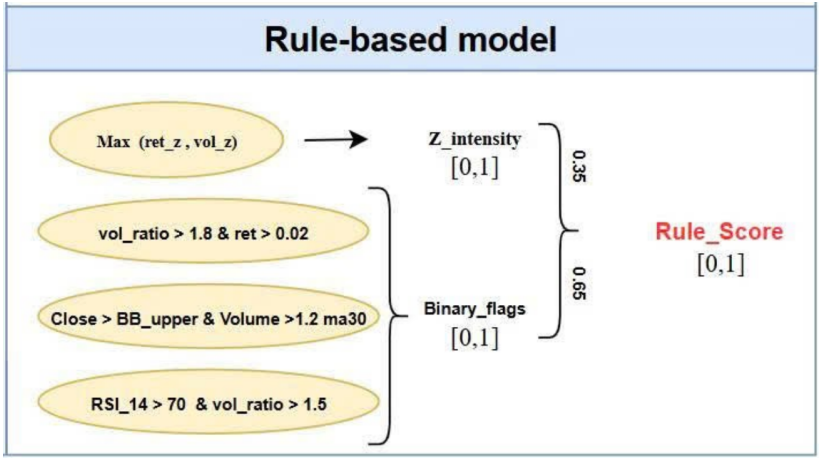
# Rule-based Model Components



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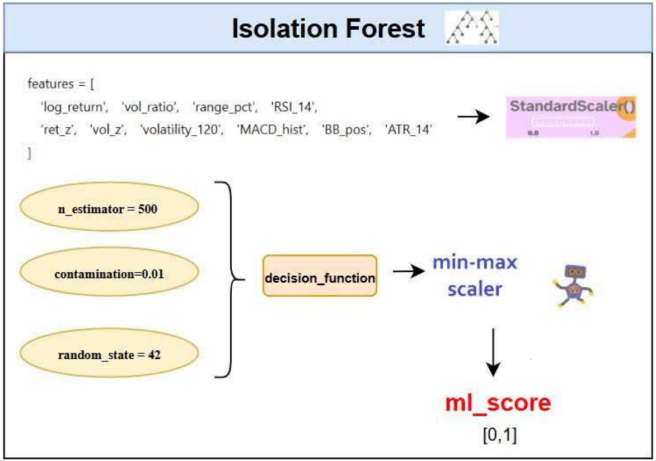
# Isolation Forest Implementation



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# Performance Evaluation

Experimental testing conducted on **1,260 trading days** (including stock codes: FLC, VNM, LKNCY).

Table: Model Performance Comparison

| **Model** | **Precision** | **Recall** | **F1** | **ROC-AUC** |
|---|---|---|---|---|
| Isolation Forest | 0.587 | 0.607 | 0.597 | 0.955 |
| Z-score | 0.562 | **0.885** | 0.688 | 0.974 |
| blue!10 **Hybrid System** | **0.700** | 0.803 | **0.748** | **0.983** |

- **Highest F1-score**: 0.748 achieved by the **Hybrid System**.
- **Processing Speed**: 32.7 $\mu$s/sample (Optimized for real-time monitoring).

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# Comparison with State-of-the-Art (SOTA) - Part 1

| Criteria | Anomaly Transformer (SOTA 2022) | MAD-GAN (SOTA 2019) | Graph Neural Networks (SOTA 2021) | OURS HYBRID SYSTEM |
|---|---|---|---|---|
| **Key Research** | Xu et al. (ICLR 2022) | Li et al. (2019) | Wang et al. (WWW 2021) | **StockSense team (2025)** |
| **Core Algorithm** | Self-Attention & Assoc. Discrepancy | Adversarial: Gen vs. Discrim | Nodes (Stocks) & Edges (Rel.) | **Ensemble: Rules + IsoForest** |
| **Data Needs** | Massive Big Data for convergence. | Large diverse sets for distribution. | Complex: News, relations, owners. | **Standard OHLCV only.** |
| **Hardware** | High: Requires V100/A100 GPUs. | High: Needs GPU for GAN training. | Mid-High: High RAM for graphs. | **Low: Runs on standard CPU.** |

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# Comparison with State-of-the-Art (SOTA) - Part 2

| Criteria | Anomaly Transformer (SOTA 2022) | MAD-GAN (SOTA 2019) | Graph Neural Networks (SOTA 2021) | OURS HYBRID SYSTEM |
|---|---|---|---|---|
| **Train/Test Time** | Train: Hours/Days.<br><br>Test: ms. | Train: Slow & unstable.<br><br>Test: Fast. | Train: Slow .<br><br>Test: Graph-dependent. | **Train: < 1s .**<br><br>**Test: fast ($32.7 \mu s$).** |
| **F1-Score** | Very High (0.85 - 0.90) | High (> 0.80) | Very High (Group fraud) | **Fair (0.748) - Reliable alerts.** |
| **Explainability** | **Low:** Black-box attention. | **Very Low:** Hidden logic. | **Medium:** Network-based. | **High:** Clear Rule/ML logic. |

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# Ideal Setting

- The system performs optimally in the context of the Vietnamese stock market or the cryptocurrency market, with a focus on mid- and small-cap stocks.
- It is designed to operate in real time on low-cost server infrastructure or directly on mobile applications.

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# Model Limitations & Future Work

## Model Limitations

- The model currently employs a fixed contamination rate (0.01) for the Isolation Forest.
- The system does not yet normalize price movements against the overall market (e.g., VNIndex), which could help filter out systemic risk.

## Future Work

- Integrating NLP techniques to analyze news sentiment related to detected anomalies.
- Exploring Autoencoders to improve feature representation.
- Conducting backtesting across a wider range of industry sectors.

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# Conclusion

- **Efficiency:** Faster and cheaper.
- **Explainability:** Explainable to users.
- **Cold Start:** Works even for newly listed stocks.
- **Maintainability:** Easy to update daily.

# THANK YOU FOR YOUR ATTENTION!

Q & A